# Exercise Set 1

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**The choice of group matters.** In the lecture, the discrete logarithm problem (DLP) was considered in an arbitrary cyclic group $\mathbb{G}$ of prime order $q$ generated by an element $g$; we write $\mathbb{G} = \langle g \rangle$ for the group generated by $g$. There, the best generic algorithms need $\mathcal{O}(\sqrt{q})$ group operations. In this exercise set we look at concrete groups in which the DLP is considerably easier, because the group has additional structure that a generic algorithm does not use. The main tools are pairings, which are also the basis of several constructions in modern cryptography. Parts marked $\star$ are intended as take-home exercises.

**Bilinear pairings:** Let $\mathbb{G}_1 = \langle P \rangle$, $\mathbb{G}_2 = \langle Q \rangle$ and $\mathbb{G}_T$ be cyclic groups of the same prime order $q$, the first two written additively and $\mathbb{G}_T$ multiplicatively. A *(bilinear) pairing* is a map

$$e : \mathbb{G}_1 \times \mathbb{G}_2 \longrightarrow \mathbb{G}_T$$

with the following three properties.

**Bilinearity** $e([a]P', [b]Q') = e(P', Q')^{ab}$ for all $P' \in \mathbb{G}_1$, $Q' \in \mathbb{G}_2$ and $a, b \in \mathbb{Z}$.
**Non-degeneracy** $e(P, Q) \neq 1$.
**Efficiency** $e$ can be evaluated in polynomial time.

If $\mathbb{G}_1 = \mathbb{G}_2$ the pairing is called *symmetric*, and we then write $\mathbb{G}$ for $\mathbb{G}_1 = \mathbb{G}_2$.

For Problems 1 and 2 no knowledge of how a pairing is constructed is needed. Problem 3 uses a concrete pairing, the Weil pairing, which is introduced there.

**Problem 1.** (Generic hardness vs. hardness)

(a) Consider the group $(\mathbb{Z}_N, +)$. Fix $g \in \mathbb{Z}_N$ with $\gcd(g, N) = 1$ and let $h = x \cdot g \bmod N$. Give an algorithm that recovers $x$ and state its cost.

**Solution.** In this group, "exponentiation" is the map $x \mapsto x \cdot g$, so the DLP is: given $g$ and $xg \bmod N$, find $x$. Since $\gcd(g, N) = 1$, one can compute $g^{-1} \bmod N$ with the extended Euclidean algorithm and output $x = h \cdot g^{-1} \bmod N$. This takes $\mathcal{O}(\log^2 N)$ bit operations, i.e. polynomial time in the bit length of $N$, compared to $\mathcal{O}(\sqrt{N})$ for the generic algorithms.

(b) Every cyclic group of order $N$ is isomorphic to $(\mathbb{Z}_N, +)$. Why does part (a) not break the DLP in *every* cyclic group of order $N$?

**Solution.** Let $\mathbb{G} = \langle g \rangle$ be a cyclic group of order $N$. To use part (a) one has to compute the isomorphism $\mathbb{G} \to \mathbb{Z}_N$, and this is the DLP itself. The map $\mathbb{Z}_N \to \mathbb{G}$, $x \mapsto g^x$, is efficient (square-and-multiply), but its inverse is the discrete logarithm map.

In other words, the hardness of the DLP depends on how the group elements are represented and how the group law is computed on these representations, not on the abstract group. The generic group model captures this by replacing the elements with random labels and the lower bound of $\Omega(\sqrt{q})$ then says nothing about algorithms that use a particular representation.

The remaining problems show elliptic curves whose representation can be exploited in a less obvious way using for example pairings.

**Problem 2.** (Pairings as a black box)

(a) Let $e : \mathbb{G} \times \mathbb{G} \to \mathbb{G}_T$ be a symmetric pairing as defined above, where $\mathbb{G} = \langle P \rangle$. On input $(P, [a]P, [b]P, [c]P)$, give an efficient test for the DDH problem, i.e. whether $c \equiv ab \pmod{q}$.

> **Solution.** Let $g_T := e(P, P)$. By non-degeneracy $g_T \neq 1$, and since $\mathbb{G}_T$ has prime order $q$, $g_T$ has order $q$. Compute $e([a]P, [b]P) = g_T^{ab}$ and $e(P, [c]P) = g_T^c$ and compare; the two values are equal iff $c \equiv ab \pmod{q}$. Both pairing evaluations are efficient by our definition.
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> Thus, in a group with an efficient pairing the decisional Diffie–Hellman problem is easy in $\mathbb{G}$, even if the DLP and the computational Diffie–Hellman problem may be hard, as given $[a]P$ and $[b]P$ one can recognise $[ab]P$ but not necessarily compute it. Groups with this property are called gap Diffie–Hellman groups.

(b) Now let $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ be a general, not necessarily symmetric, pairing as defined above, with $\mathbb{G}_1 = \langle P \rangle$ and $\mathbb{G}_2 = \langle Q \rangle$. Let $R = [x]P \in \mathbb{G}_1$ for unknown $x$. Construct in polynomial time an instance of the DLP in $\mathbb{G}_T$ whose solution is $x$. What does this say about the hardness of the DLP in $\mathbb{G}_1$ versus $\mathbb{G}_T$?

> **Solution.** Set $\alpha := e(P, Q)$ and $\beta := e(R, Q)$. By bilinearity, $\beta = e([x]P, Q) = \alpha^x$, and $\alpha$ has order $q$ as in part (a), so $(\alpha, \beta)$ is a DLP instance in $\mathbb{G}_T$ with solution $x \bmod q$. Therefore
>
> $$\text{time(DLP in } \mathbb{G}_1) \;\leq\; \text{time(DLP in } \mathbb{G}_T) + \text{poly},$$
>
> i.e. the DLP in $\mathbb{G}_1$ is at most as hard as the DLP in $\mathbb{G}_T$. Comparing with Problem 1(b) we notice that here the isomorphism from $\mathbb{G}_1$ to $\mathbb{G}_T$ can be computed efficiently.

(c) $\star$ Given a group $\mathbb{G} = \langle P \rangle$ with a symmetric pairing, design a one-round three-party key exchange. That is, how do Alice, Bob and Charlie holding secrets $a, b, c$ agree on a shared secret key if each can only broadcast *one* message?

> **Solution.** Alice, Bob and Charlie broadcast $[a]P$, $[b]P$ and $[c]P$. Alice computes $e([b]P, [c]P)^a = e(P, P)^{abc}$, and Bob and Charlie proceed analogously, so all three obtain $K = e(P, P)^{abc}$. Without a pairing, three-party Diffie–Hellman requires two rounds. The underlying assumption is the bilinear Diffie–Hellman problem: given $[a]P, [b]P, [c]P$, compute $e(P, P)^{abc}$.

(d) $\star$ Let $H : \{0,1\}^* \to \mathbb{G}_1$ be a hash function onto $\mathbb{G}_1$. The secret key is $x$, the public key $\text{pk} = [x]Q \in \mathbb{G}_2$, and the signature on $m$ is $\sigma = [x]H(m) \in \mathbb{G}_1$. Give the verification equation and prove correctness.

> **Solution.** Accept iff $e(\sigma, Q) = e(H(m), \text{pk})$. For a correctly generated signature,
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> $$e(\sigma, Q) = e([x]H(m), Q) = e(H(m), Q)^x = e(H(m), [x]Q) = e(H(m), \text{pk}).$$
>
> A signature is a single element of $\mathbb{G}_1$ (48 bytes for BLS12-381 in compressed form). Further, signatures on the same message can be aggregated meaning that their sum verifies against the sum of the public keys.

(e) In all pairings we know how to build, $\mathbb{G}_1$ is a subgroup of an elliptic curve group $E(\mathbb{F}_p)$ and $\mathbb{G}_T$ is a subgroup of $\mathbb{F}_{p^k}^*$ for some $k \geq 1$. The best known algorithm for the DLP

in $\mathbb{G}_1$ is Pollard's rho method with $\mathcal{O}(\sqrt{q})$ group operations, while the DLP in $\mathbb{F}_{p^k}^*$ can be solved with the Number Field Sieve in subexponential time $L_{p^k}(1/3)$. Here $\mathbb{F}_{p^k}^*$ is the multiplicative group of the finite field with $p^k$ elements, $k$ is called the *embedding degree* (defined before Problem 3), and $L_{p^k}(1/3) = \exp\left(c\,(\ln p^k)^{1/3}(\ln\ln p^k)^{2/3}\right)$ for a constant $c$, a function that grows slower than any exponential in $\log p^k$. Given part (b), what does this mean for the choice of parameters if we *want* to use a pairing?

**Solution.** By part (b), the DLP in $\mathbb{G}_1$ is at most as hard as the DLP in $\mathbb{G}_T \subset \mathbb{F}_{p^k}^*$, so both groups have to be secure. For $\lambda$ bits of security one needs $q \geq 2^{2\lambda}$ because of Pollard rho, and $p^k$ large enough that the NFS in $\mathbb{F}_{p^k}$ costs about $2^\lambda$. Since the NFS is subexponential, the field has to be several thousand bits large, while the curve only needs a few hundred bits. The embedding degree $k$ has to be chosen with some care: small enough that arithmetic in $\mathbb{F}_{p^k}$ is fast, large enough that the NFS in $\mathbb{F}_{p^k}$ is infeasible. For example, BLS12-381 (used in Ethereum and for BLS signatures) has $\log_2 p = 381$ and $k = 12$, so $\mathbb{G}_T \subset \mathbb{F}_{p^{12}}$ with 4572 bits.

**Background for Problem 3.** Let $E$ be an elliptic curve over $\mathbb{F}_p$ and $q \neq p$ a prime. $\#E(\mathbb{F}_p)$ denotes the number of points of $E$ with coordinates in $\mathbb{F}_p$ (including $\mathcal{O}$), and $\overline{\mathbb{F}}_p$ the algebraic closure of $\mathbb{F}_p$, i.e. the union of all finite extensions $\mathbb{F}_{p^j}$; a point of $E(\overline{\mathbb{F}}_p)$ has coordinates in some $\mathbb{F}_{p^j}$. The following facts may be used without proof.

**Counting** $\#E(\mathbb{F}_p)$ can be computed in polynomial time (e.g. SEA algorithm), and $\#E(\mathbb{F}_p)$ determines $\#E(\mathbb{F}_{p^k})$ for every $k$ by a closed formula. Random points of $E(\mathbb{F}_{p^k})$ can be sampled efficiently.

**Extensions** $E(\mathbb{F}_p) \subseteq E(\mathbb{F}_{p^k})$ for every $k \geq 1$, with the same addition formulas. $\mathbb{F}_{p^k}^*$ is cyclic of order $p^k - 1$; a cyclic group of order $M$ has a subgroup of order $q$ iff $q \mid M$.

**Torsion** The $q$-torsion $E[q] := \{S \in E(\overline{\mathbb{F}}_p) : [q]S = \mathcal{O}\}$ is a group isomorphic to $\mathbb{Z}_q \times \mathbb{Z}_q$ with $q^2$ points; a point $P \in E(\mathbb{F}_p)$ of order $q$ generates only one of its $q+1$ subgroups of order $q$. The *embedding degree* $k$ is the smallest integer with $q \mid p^k - 1$. If $q \nmid p - 1$, then $E[q] \subseteq E(\mathbb{F}_{p^k})$.

**Weil pairing** There is a map $e_q : E[q] \times E[q] \to \mu_q := \{\zeta \in \overline{\mathbb{F}}_p : \zeta^q = 1\}$ that is bilinear, *alternating* ($e_q(S,S) = 1$ for all $S$), non-degenerate, and computable by Miller's algorithm in time polynomial in $\log p^k$.

**Problem 3.** (The MOV attack)

We now apply Problem 2(b) to build an attack on the DLP for certain elliptic curves. This attack is due to Menezes, Okamoto and Vanstone (1991).

Let $P \in E(\mathbb{F}_p)$ be a point of prime order $q \neq p$ with $q \nmid p - 1$, let $R = [x]P$, and let $k$ be the curve's embedding degree. The pairing will be $e_q$ with $\mathbb{G}_1 = \langle P \rangle$ and $\mathbb{G}_T = \mu_q$.

(a) Show that $\mu_q \subseteq \mathbb{F}_{p^k}^*$ and that $k$ is minimal with this property. Conclude that all points of $E[q]$ and all pairing values live in $\mathbb{F}_{p^k}$.

**Solution.** Since $\mathbb{F}_{p^k}^*$ is cyclic of order $p^k - 1$, it has a subgroup of order $q$ iff $q \mid p^k - 1$. Such a subgroup consists of $q$ elements $\zeta$ with $\zeta^q = 1$; as the polynomial $X^q - 1$ has at most $q$ roots in a field, this subgroup is $\mu_q$. Hence $\mu_q \subseteq \mathbb{F}_{p^k}^*$ iff $q \mid p^k - 1$.

Conversely, if $\mu_q \subseteq \mathbb{F}_{p^j}^*$ for some $j$, then $q \mid p^j - 1$ by Lagrange's theorem. So the smallest such $j$ is the embedding degree $k$.

Since $q \nmid p-1$, the background information lets us deduce $E[q] \subseteq E(\mathbb{F}_{p^k})$. Thus both the arguments and the values of $e_q$ have coordinates in $\mathbb{F}_{p^k}$. This is the field in which the attack computes, and its size $k \log_2 p$ determines the cost.

(b) Fix $T \in E[q]$ and define $\varphi_T : \langle P \rangle \to \mu_q$, $S \mapsto e_q(S, T)$. Show that $\varphi_T$ is a group homomorphism, and determine exactly when it is injective.

**Solution.** By bilinearity in the first argument,

$$\varphi_T(S + S') = e_q(S + S', T) = e_q(S, T)\, e_q(S', T) = \varphi_T(S)\, \varphi_T(S'),$$

so $\varphi_T$ is a homomorphism. In particular $\varphi_T([c]P) = e_q(P, T)^c$ for all $c$.

A homomorphism is injective iff its kernel is trivial. The kernel of $\varphi_T$ is a subgroup of $\langle P \rangle$, a group of prime order $q$, so by Lagrange's theorem it is either $\{\mathcal{O}\}$ or $\langle P \rangle$. It equals $\langle P \rangle$ iff it contains the generator $P$, i.e. iff $e_q(P, T) = 1$. Hence

$$\varphi_T \text{ is injective } \iff e_q(P, T) \neq 1.$$

The map from the solution of Problem 2(b) is $\varphi_Q$.

(c) By part (b), we want a point $T \in E[q]$ with $e_q(P, T) \neq 1$, so that $\varphi_T$ is injective. Show that every $T \in \langle P \rangle$ is a bad choice. Then compute the probability that a uniformly random $T \in E[q]$ is bad.

**Solution.** Let $T = [c]P \in \langle P \rangle$. By bilinearity in the second argument and the alternating property, $e_q(P, [c]P) = e_q(P, P)^c = 1$, so $T$ is a bad choice. Hence $T$ has to be chosen outside $\langle P \rangle$. In general, $E(\mathbb{F}_p)$ contains no points of order $q$ outside $\langle P \rangle$, which is why the attack works in $E(\mathbb{F}_{p^k})$, where all of $E[q]$ is available.

For the probability, consider $\psi : E[q] \to \mu_q$, $T \mapsto e_q(P, T)$, which is a homomorphism by bilinearity in the second argument. Its kernel $\ker \psi = \{T \in E[q] : e_q(P, T) = 1\}$ is exactly the set of bad choices.

- $\psi$ is not trivial: otherwise $e_q(P, T) = 1$ for all $T \in E[q]$, and non-degeneracy would give $P = \mathcal{O}$.
- The image of $\psi$ is a non-trivial subgroup of $\mu_q$, which has prime order $q$, so $\psi$ is surjective and $|\ker \psi| = |E[q]|/|\mu_q| = q^2/q = q$.
- $\ker \psi \supseteq \langle P \rangle$, since we showed above that $e_q(P, T) = 1$ for all $T \in \langle P \rangle$. Both sets have $q$ elements, so $\ker \psi = \langle P \rangle$.

So the bad choices are exactly the points of $\langle P \rangle$, and a uniformly random $T \in E[q]$ is bad with probability $|\langle P \rangle|/|E[q]| = q/q^2 = 1/q$. A bad choice is also easy to detect, since one can check $e_q(P, T) \neq 1$ efficiently.

(d) Let $N := \#E(\mathbb{F}_{p^k})$. Explain why $q^2 \mid N$. Assume $q^3 \nmid N$ and let $T' \in E(\mathbb{F}_{p^k})$ be uniformly random. Show that $T := [N/q^2]\, T'$ is uniformly random in $E[q]$. Why does $[N/q]\, T'$ not work?

**Solution.** By (a), $E[q]$ is a subgroup of $E(\mathbb{F}_{p^k})$ with $q^2$ elements, so $q^2 \mid N$ by Lagrange's theorem. Write $N = q^2 m$; the assumption $q^3 \nmid N$ means $\gcd(m, q) = 1$.

Note that $[m]T' \in E[q]$ for every $T'$ and let $S := [m]T'$. By (a), $E[q] \subseteq E(\mathbb{F}_{p^k})$, and $E[q]$ has $q^2$ elements (background box). By Lagrange's theorem $[q^2]S = [N]T' = \mathcal{O}$,

so the order of $S$ divides $q^2$. If $S$ had order $q^2$, the subgroup generated by $S$ and $E[q]$ would have $|\langle S\rangle| \cdot |E[q]|/|\langle S\rangle \cap E[q]| = q^2 \cdot q^2/q = q^3$ elements, so $q^3 \mid N$ by Lagrange's theorem, a contradiction. Hence $[q]S = \mathcal{O}$.

For uniformity, decompose the group $G = E(\mathbb{F}_{p^k})$ as $G = G_q \times G'$, where $G_q$ consists of the elements of order a power of $q$ and $G'$ of the elements of order coprime to $q$, so $|G_q| = q^2$ and $|G'| = m$. Since $E[q] \subseteq G_q$ and both have $q^2$ elements, $G_q = E[q]$. As $m = |G'|$, multiplication by $m$ is zero on $G'$. Further, it is a bijection on $E[q]$ (as $\gcd(m, q) = 1$). Hence $T' \mapsto [m]T'$ is a surjection $G \to E[q]$ with kernel $G'$, every element of $E[q]$ has exactly $m$ preimages, and a uniform $T'$ gives a uniform $T \in E[q]$.

Multiplying by $N/q = qm$ instead gives $[qm]T' = [q]([m]T') = \mathcal{O}$ for every $T'$, since $[m]T' \in E[q]$.

(e) Assemble (a)–(d) into an algorithm that, on input $(E, P, R)$, reduces the ECDLP in $\langle P\rangle$ to a DLP in $\mathbb{F}_{p^k}^*$ and then solves the latter with the Number Field Sieve. State the cost of each step. For $\log_2 p = 256$ and $q \approx p$, for which $k$ is this faster than Pollard rho on the curve? You may use that the NFS in a finite field of about 3072 bits costs roughly $2^{128}$ operations.

**Solution. Input:** $E/\mathbb{F}_p$, $P \in E(\mathbb{F}_p)$ of prime order $q$, $R = [x]P$. **Output:** $x \in \mathbb{Z}_q$.

1. Compute $p^j \bmod q$ for $j = 1, 2, \ldots$ until the result is 1; this $j$ is $k$. Stop if $k$ exceeds a fixed bound.
2. Construct $\mathbb{F}_{p^k}$ and regard $P, R$ as points of $E(\mathbb{F}_{p^k})$. Compute $N = \#E(\mathbb{F}_{p^k})$ from $\#E(\mathbb{F}_p)$ and set $m := N/q^2$.
3. Sample $T' \in E(\mathbb{F}_{p^k})$ uniformly at random and set $T := [m]T'$, as in (d).
4. Compute $\alpha := e_q(P, T)$. If $\alpha = 1$ (probability $1/q$ by (c)), return to step 3.
5. Compute $\beta := e_q(R, T)$. By bilinearity, $\beta = e_q(P, T)^x = \alpha^x$.
6. Solve the DLP $\beta = \alpha^x$ in $\mathbb{F}_{p^k}^*$ with the Number Field Sieve and output $x$.

Steps 1–5 take polynomial time in $\log p^k$: step 1 is at most $k$ multiplications modulo $q$, step 2 is polynomial by the counting fact, step 3 is one scalar multiplication, and steps 4 and 5 are two evaluations of Miller's algorithm. Step 6 dominates, so

$$\text{time(MOV)} \;=\; L_{p^k}(1/3) + \text{poly}(\log p^k) \qquad \text{versus} \qquad \text{time(rho)} \;=\; \mathcal{O}(\sqrt{q}).$$

The attack is faster than Pollard rho iff the NFS in $\mathbb{F}_{p^k}$ costs less than $\sqrt{q} \approx 2^{128}$, i.e. iff $\mathbb{F}_{p^k}$ has fewer than about 3072 bits. With $\log_2 p = 256$ this means $256\,k < 3072$, i.e. $k \le 11$. The larger $k$, the larger the field and the slower the NFS, so the attack is most damaging for $k = 1, 2$ and becomes irrelevant beyond $k \approx 12$.